

Chapter 4

The Medium Access Control Sublayer

1. THE CHANNEL ALLOCATION
PROBLEM
2. MULTIPLE ACCESS PROTOCOLS
3. ETHERNET

Networks can be divided into two categories:

- 1) Those using **point-to-point** connections
and
- 2) Those using **broadcast** channels.

Here we deal with **broadcast** networks and their protocols.

In any **broadcast network**, the key issue is how to determine who gets to use the channel when there is competition for it.

To make this point clearer, consider a conference call in which six ppeoples, on six different telephones, are all connected together so that each one can hear and talk to all the others.

It is very likely that when one of them stops speaking, two or more will start talking at once, leading to chaos.

In a face-to-face meeting, chaos is avoided by external means, for example, at a meeting, people raise their hands to request permission to speak.

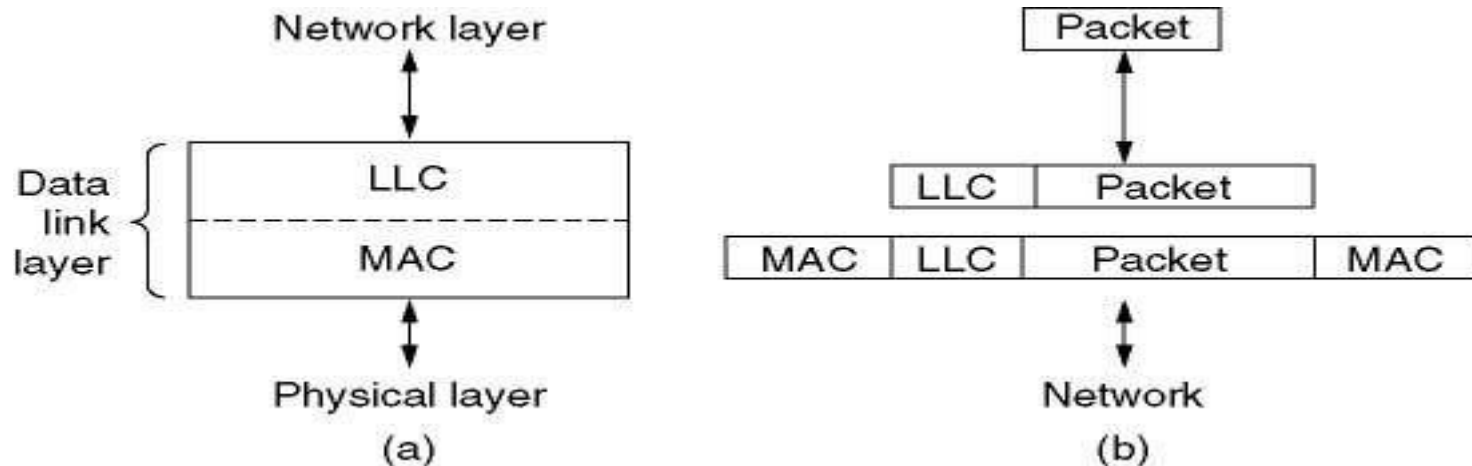
When only a single channel is available, determining **who should go next** is much harder.

Many protocols for solving the problem are known and form the contents of this ppt.

In the literature, broadcast channels are sometimes referred to as **multi-access channels** or **random access channels**

The protocols used to determine who goes next on a multi-access channel belong to a sublayer of the data link layer called the MAC (**Medium Access Control**) sublayer.

Technically, the MAC sublayer is the bottom part of **the data link layer**.



The MAC sublayer is especially important in **LANs**, nearly all of which use a multi-access channel as the basis of their communication.

WANs, in contrast, use point-to-point links, except for satellite networks.

Because multi-access channels and LANs are so closely related, in this chapter we will discuss **LANs in general**, as well as **satellite and some other broadcast networks**

4.1. THE CHANNEL ALLOCATION PROBLEM

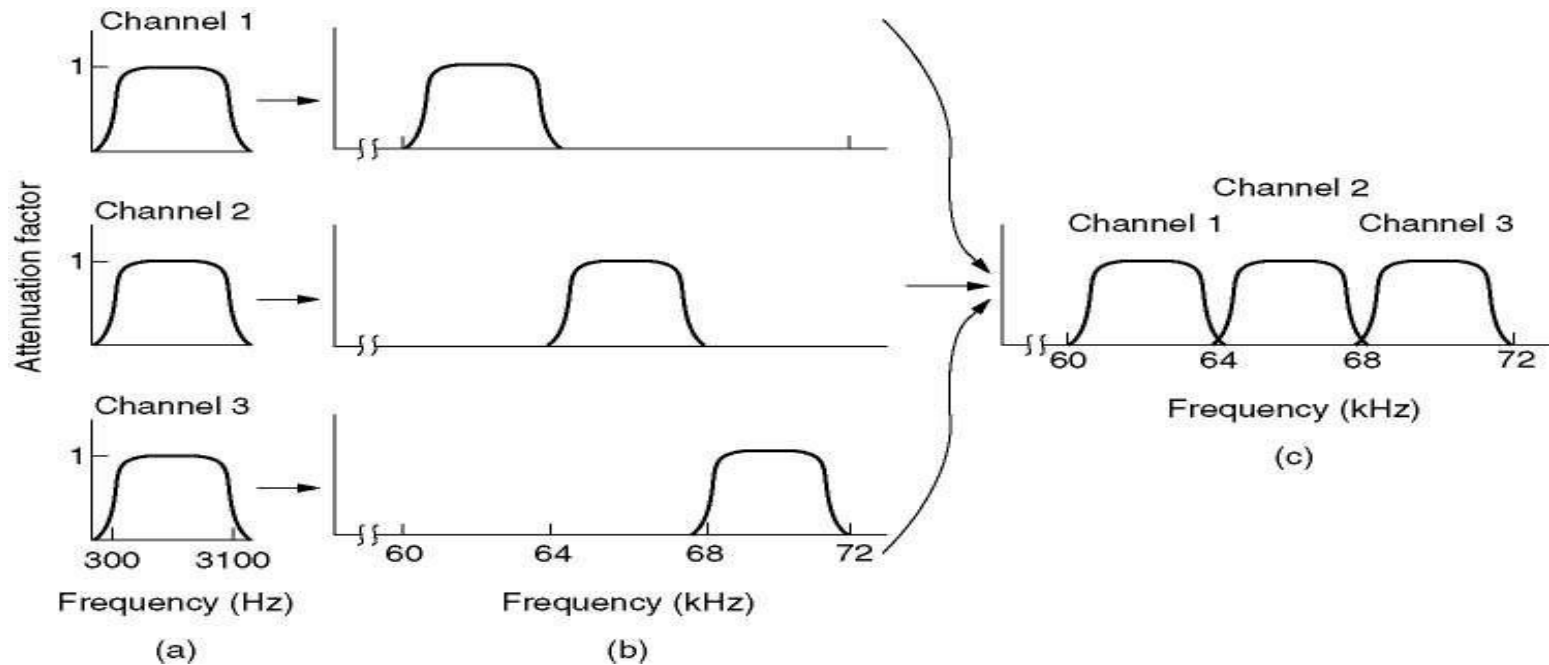
The central theme of this chapter is how to allocate a single broadcast channel among competing users.

We will first look at static and dynamic schemes in general. Then we will examine a number of specific algorithms.

4.1.1.Static Channel Allocation in LANs and MANs

The traditional way of allocation a single channel, such as a telephone trunk, among multiple competing users is **Frequency Division Multiplexing** (FDM). If there are **N** users, the bandwidth is divided into **N** equal sized portions, each user being assigned one portion.

Frequency Division Multiplexing



- (a) The original bandwidths.
- (b) The bandwidths raised in frequency.
- (b) The multiplexed channel.

Since each user has a private frequency band, there is no interference between users

When there is only a small and fixed number of users, each of which has a heavy (buffered) load of traffic (e.g., carriers' switching offices), FDM is a simple and efficient allocation mechanism.

However, when the number of senders is large and continuously varying, or the traffic is bursty, FDM presents some problems.

If the spectrum is cut up into N regions, and fewer than N users are currently interested in communicating, a large piece of valuable spectrum will be wasted.

If more than N users want to communicate, some of them will be denied permission, for lack of bandwidth, even if some of the users who have been assigned a frequency band hardly ever transmit or receive anything.

However, even assuming that the number of users could somehow be held constant at N , dividing the single available channel into static subchannels is inherently inefficient.

The basic problem is that when some users are quiescent, their bandwidth is simply lost. They are not using it, and no one else is allowed to use it either.

Furthermore, in most computer systems, data traffic is extremely bursty (peak traffic to mean traffic ratios of 1000:1 are common).

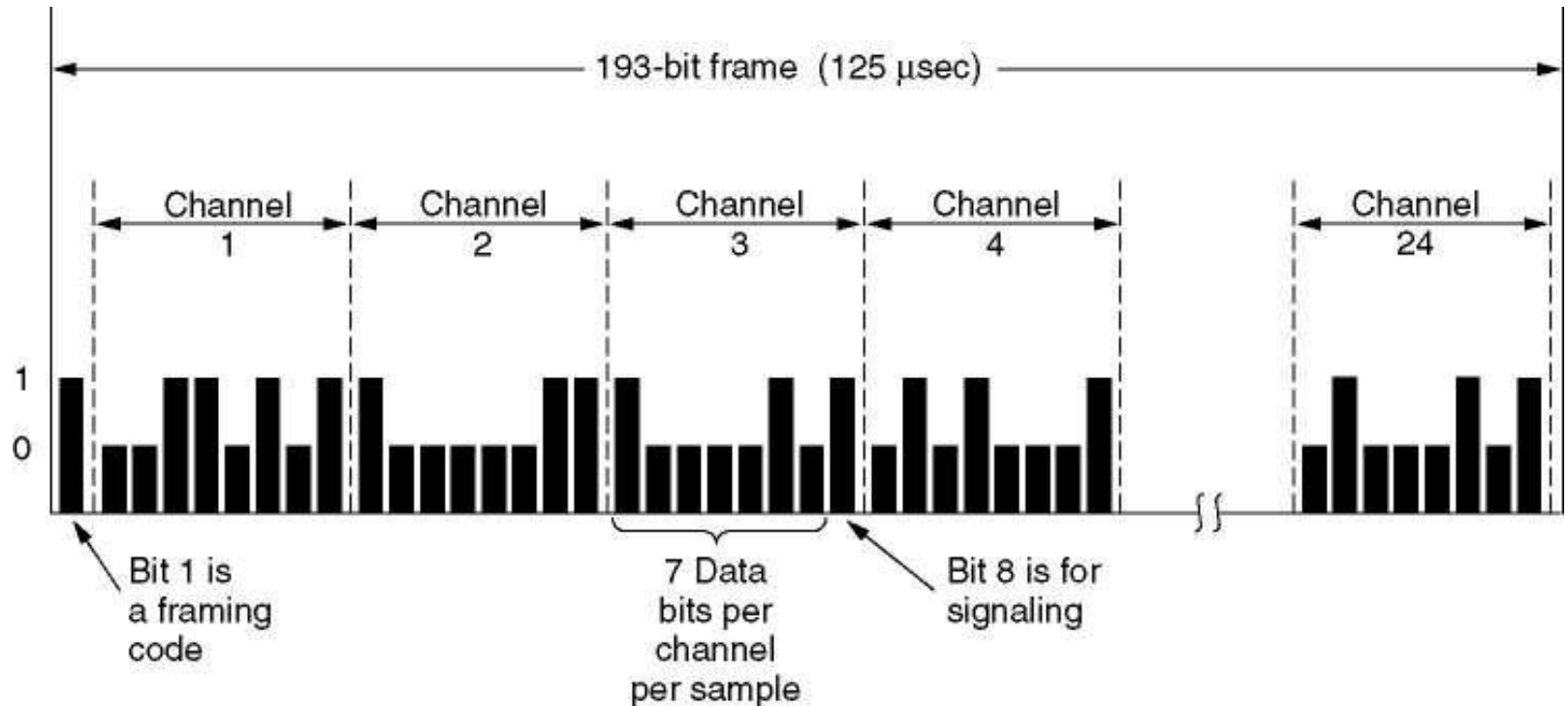
Consequently, most of the channels will be idle most of the time.

Precisely the same arguments that apply to FDM also apply to time division multiplexing (TDM).

Each user is statically allocated every N th time slot.

If a user does not use the allocated slot, it just lies fallow.

Time Division Multiplexing



The T1 carrier (1.544 Mbps).

Since none of the traditional static channel allocation methods work well with bursty traffic, we will now explore dynamic methods.

4.1.2. Dynamic Channel Allocation in LANs and MANs

Before we get into the first of the many channel allocation methods to be discussed here, it is worthwhile carefully formulating the allocation problem.

Underlying all the work done in this area are five key assumptions, described below.

1. Station Model.

The model consists of **N** independent **stations** (e.g., computers, telephones, or personnel communicators), each with a program or user that generates frames for transmissions. Stations are sometimes called **terminals**.

Once a frame has been generated, the station is blocked and does nothing until the frame has been successfully transmitted.

2. Single Channel Assumption.

A single channel is available for all communication. All stations can transmit on it and all can receive from it.

3. Collision Assumption.

If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a **collision**. All stations can detect collisions.

4.(a) Continuous Time.

Frame transmission can begin at any instant. There is no master clock dividing time into discrete intervals.

(b) Slotted Time.

Time is divided into discrete intervals (slots). Frame transmissions always begin at the start of a slot. A slot may contain 0, 1, or more frames, corresponding to an idle slot, a successful transmission, or a collision, respectively.

5.(a) Carrier Sense.

Stations can tell if the channel is in use before trying to use it. If the channel is sensed as busy, no station will attempt to use it until it goes idle.

(b) No Carrier Sense.

Stations cannot sense the channel before trying to use it. They go ahead and transmit. Only later can determine whether the transmission was successful.

4.2. Multiple Access Protocols

Many Algorithms for Allocating a Multiple Access Channel are Known.

Method	Description
FDM	Dedicate a frequency band to each station
WDM	A dynamic FDM scheme for fiber
TDM	Dedicate a time slot to each station
Pure ALOHA	Unsynchronized transmission at any instant
Slotted ALOHA	Random transmission in well-defined time slots
1-persistent CSMA	Standard carrier sense multiple access
Nonpersistent CSMA	Random delay when channel is sensed busy
P-persistent CSMA	CSMA, but with a probability of p of persisting
CSMA/CD	CSMA, but abort on detecting a collision
Bit map	Round robin scheduling using a bit map
Binary countdown	Highest numbered ready station goes next
Tree walk	Reduced contention by selective enabling
MACA, MACAW	Wireless LAN protocols
Ethernet	CSMA/CD with binary exponential backoff
FHSS	Frequency hopping spread spectrum
DSSS	Direct sequence spread spectrum
CSMA/CA	Carrier sense multiple access with collision avoidance

Channel allocation methods and systems for a common channel.

- ALOHA
- Carrier Sense Multiple Access Protocols
(In Syllabus)
- Collision-Free Protocols
- Limited-Contention Protocols
- Wavelength Division Multiple Access
Protocols
- Wireless LAN Protocols

ALOHA (Not in Syllabus)

Although Abramson's work, called ALOHA system, used ground-based radio broadcasting, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel.

We will discuss two versions of ALOHA here: **pure** (1970) and **slotted** (1972). They differ with respect to whether time is divided into discrete slots into which all frames must fit. Pure ALOHA does not require global time synchronization; slotted ALOHA does.

Pure ALOHA (1)

The basic idea of an ALOHA system is simple: let users transmit whenever they have data to be sent.

There will be collisions, of course, and the colliding frames will be damaged.

However, due to the feedback property of broadcasting, a sender can always find out whether its frame was destroyed by listening to the channel, the same way other users do.

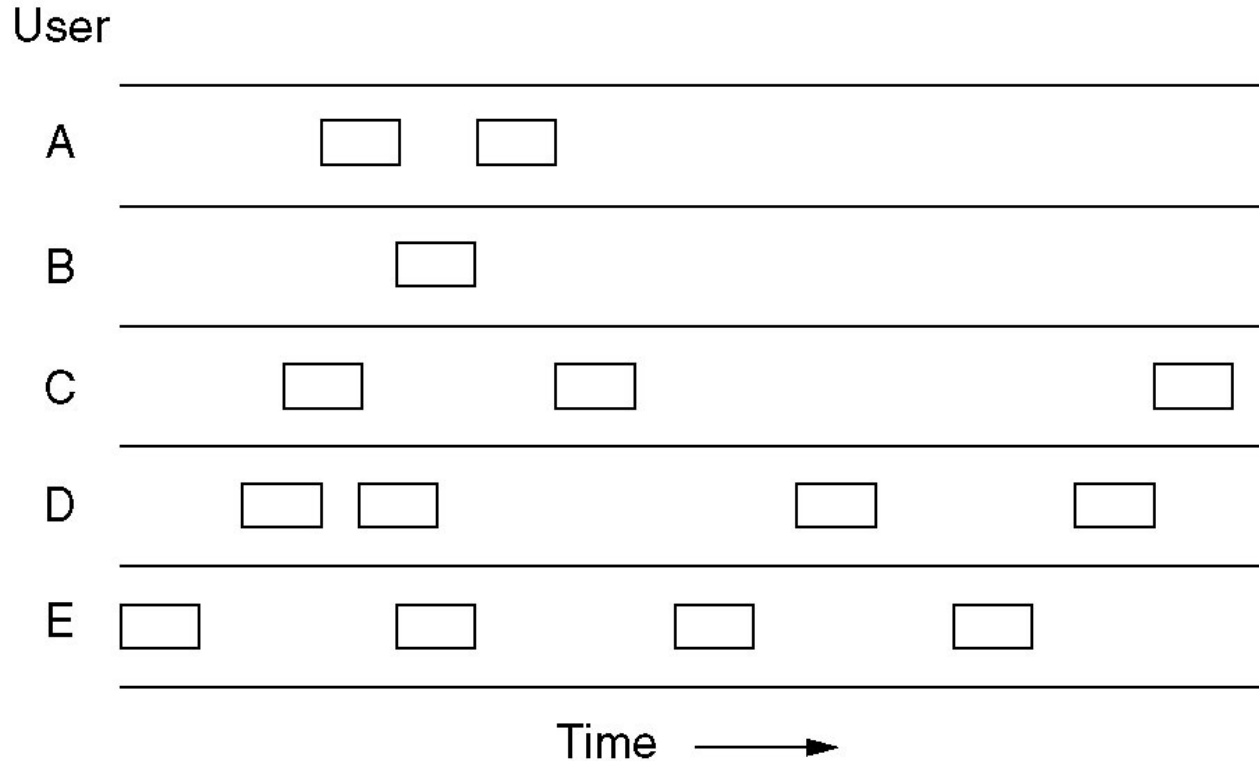
Pure ALOHA (2)

If listening while transmitting is not possible for some reason, acknowledgements are needed.

If the frame was destroyed, the sender just waits a random amount of time and sends it again.

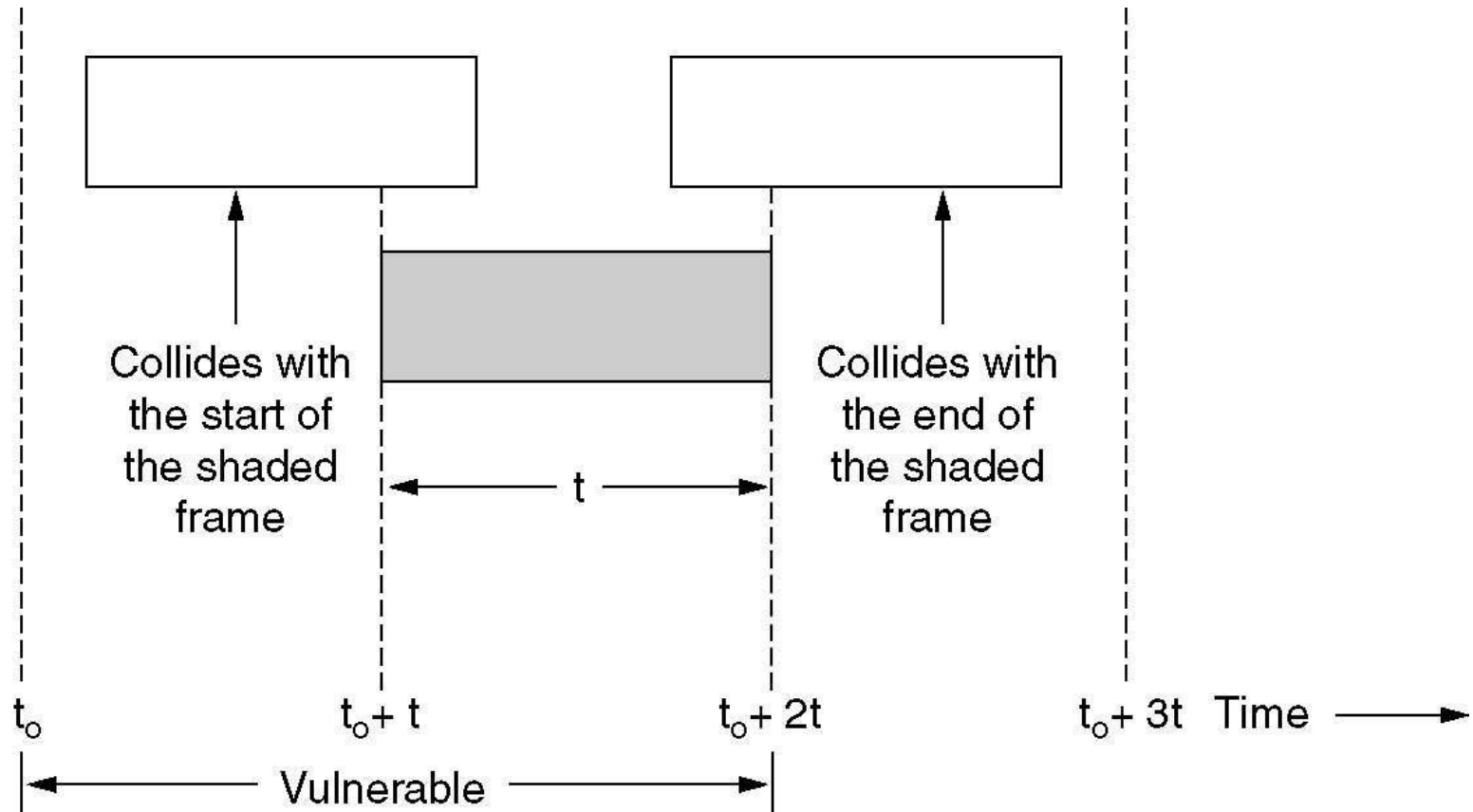
Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as **contention systems**.

Pure ALOHA (3)



In pure ALOHA, frames are transmitted at completely arbitrary times.

Pure ALOHA (4)

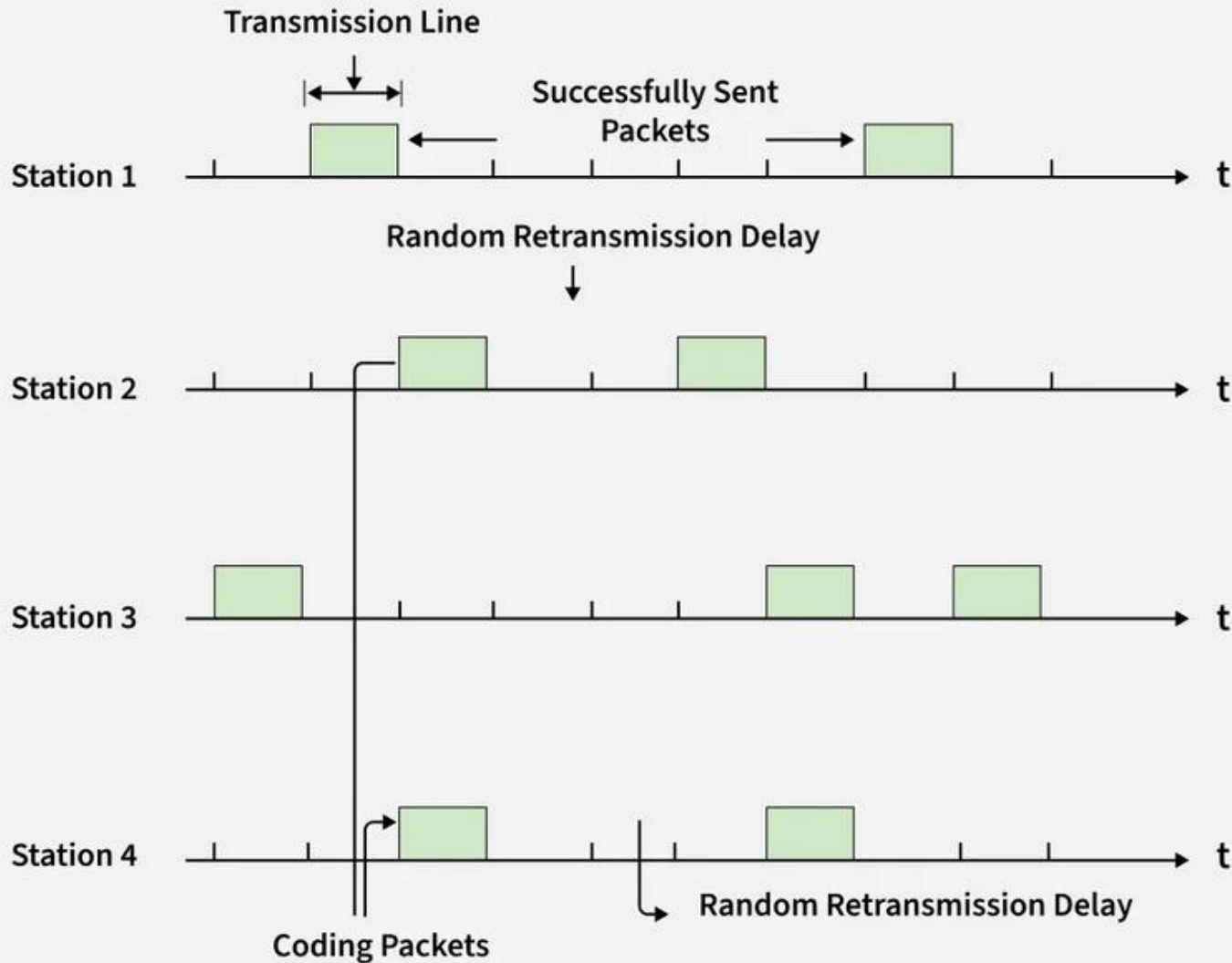


Vulnerable period for the shaded frame.

Slotted ALOHA (1)

Slotted ALOHA is an improved version of the Pure ALOHA protocol that was introduced to increase network efficiency and reduce collisions. The main idea behind Slotted ALOHA is to divide the channel into fixed-length time slots, where each station is allowed to transmit only at the beginning of a time slot.

Slotted ALOHA (2)



Slotted ALOHA (3)

Applications of Slotted ALOHA

- While rarely used in modern large-scale networks, Slotted ALOHA is still applied in:
 - Satellite communication systems
 - Wireless sensor networks
 - RFID tag communication systems
- As a foundation for CSMA (Carrier Sense Multiple Access) protocols in Ethernet and Wi-Fi.

Slotted ALOHA (3)

Advantages of Slotted ALOHA

- Higher Efficiency: Maximum throughput of 36.8%, double that of Pure ALOHA.
- Simple Implementation: Easy to design for systems with moderate traffic.
- Better Collision Control: Synchronization reduces collision probability.
- Low Overhead: Does not require heavy coordination or complex mechanisms.

Disadvantages of Slotted ALOHA

- Synchronization Overhead: Requires clock synchronization among nodes.
- Collisions Still Occur: If two or more devices transmit in the same slot, collisions cannot be avoided.
- Low Throughput Compared to Modern Protocols: Even at best, efficiency is only 36.8%.
- Increased Delay: Nodes must wait until the next slot to transmit, causing extra delay compared to Pure ALOHA.

4.2.2. Carrier Sense Multiple Access Protocols (In our syllabus)

With slotted ALOHA the best channel utilization that can be achieved is $1/e = 0.368$. This is hardly surprising, since with stations transmitting at will, without paying attention to what the other station are doing, there are bound to be many collisions.

Carrier Sense Multiple Access Protocols (2)

In local area networks, however, it is possible for station to detect what other station are doing, and adapt their behavior accordingly. This networks can achieve a much better utilization than $1/e$.

In this section we will discuss some protocols for improving performance

Carrier Sense Multiple Access Protocols (3)

*Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called **carrier sense protocols**.*

Carrier Sense Multiple Access (CSMA) is a method used in computer networks to help devices share a communication channel without interfering with each other. Before a device sends data, it listens to the channel (or senses the carrier) to check if it's free.

- If the channel is busy, the device waits until it becomes idle.
- This helps prevent data collisions, which can happen when two devices try to send data at the same time.
- CSMA is widely used in networking technologies like Ethernet and Wi-Fi.

Types of CSMA Protocol

1. CSMA/CA

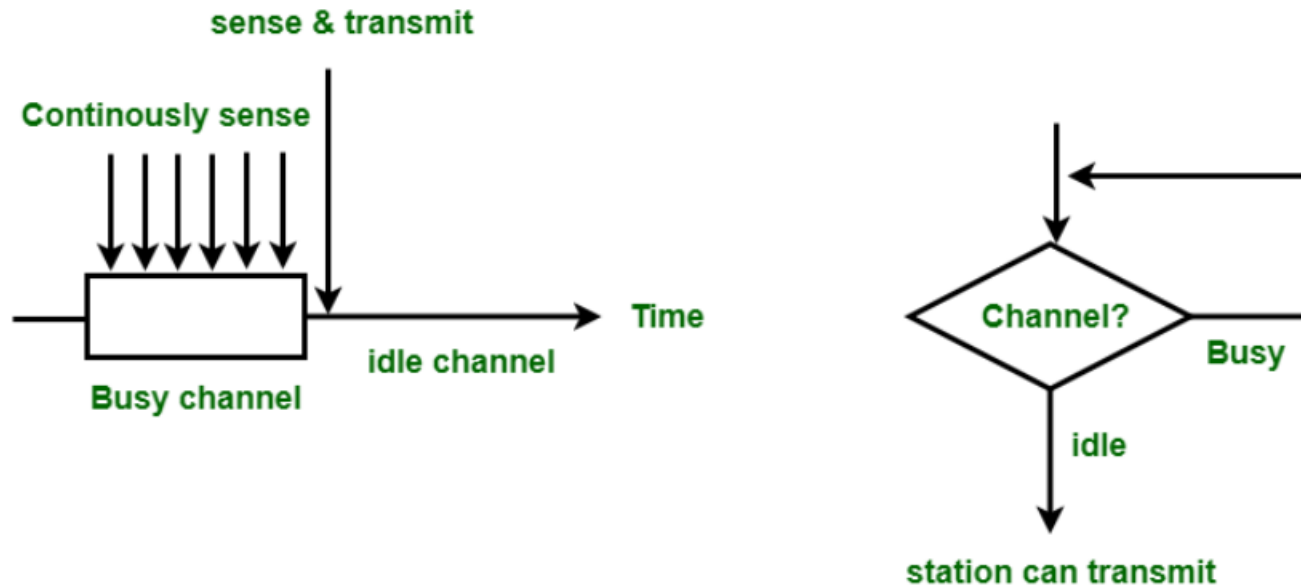
- a. 1-persistent CSMA (Carrier Sense Multiple Access)
- b. nonpersistent CSMA
- c. p-persistent CSMA

2. CSMA with Collision Detection (CSMA/CD)

1-persistent CSMA (Carrier Sense Multiple Access)

When a station has data to send, it first listens to the channel to see if anyone else is transmitting at that moment. If the channel is busy, the station waits until it becomes idle. When the station detects an idle channel, it transmits a frame. If a collision occurs, the station waits a random amount of time and starts all over again. The protocol is called 1-persistent because the station transmits with a probability of 1 when it finds the channel idle.

1-persistent CSMA (Carrier Sense Multiple Access)



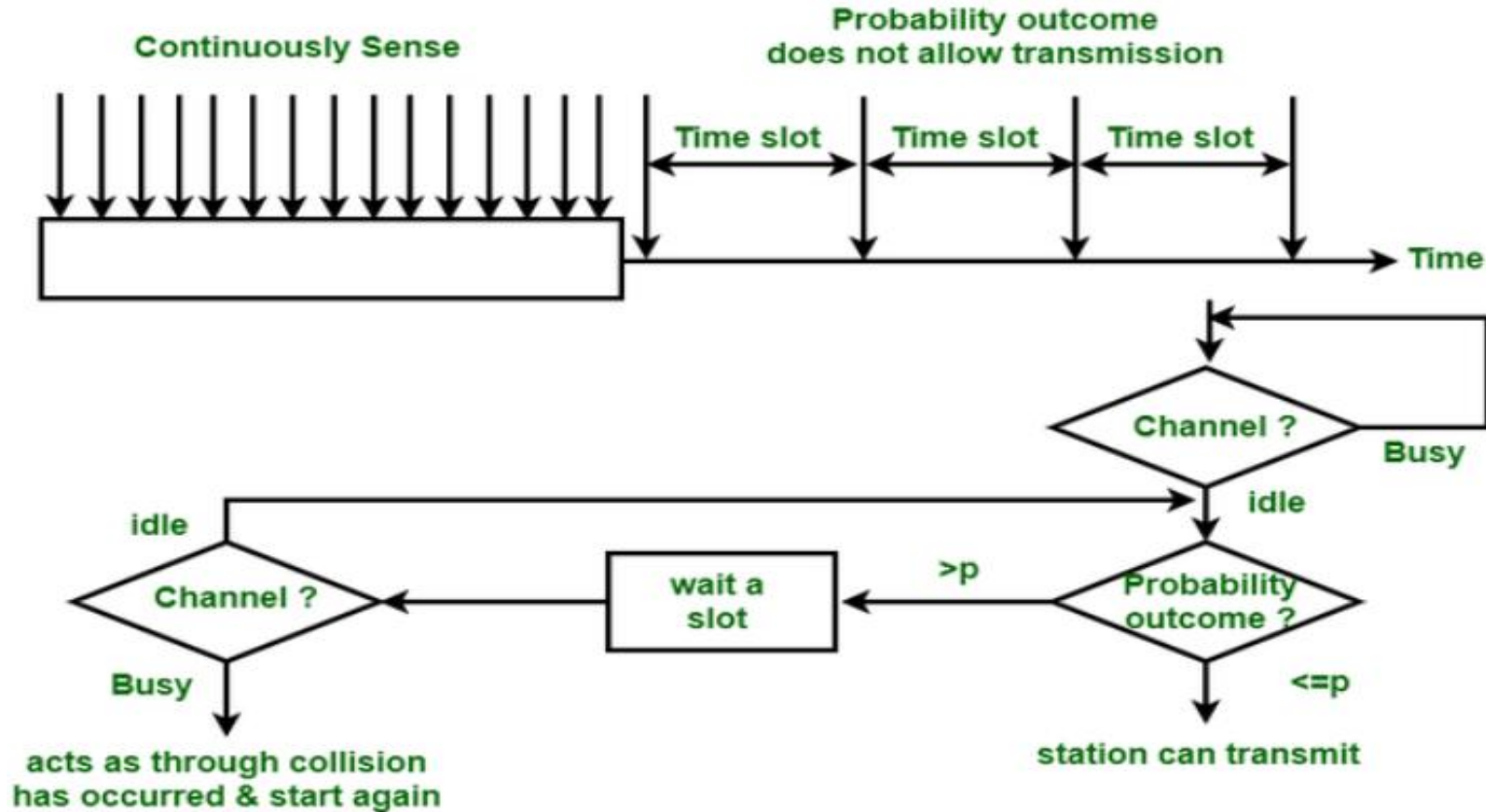
Key Points:

- High risk of collision: If multiple stations detect the channel is idle at the same time, they all transmit at once, causing a collision.
- After a collision: Each station waits for a random backoff time, then tries again once the channel is idle.
- Additional risk: If two stations are already communicating, other stations (called intermediate stations) will wait and not transmit.

Nonpersistent CSMA (Carrier Sense Multiple Access)

Before sending, a station senses the channel. If no one else is sending, the station begins doing so itself. However, if the channel is already in use, the station does not continually sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission. Instead, it waits a random period of time and then repeats the algorithm. Consequently, this algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

Nonpersistent CSMA (Carrier Sense Multiple Access)



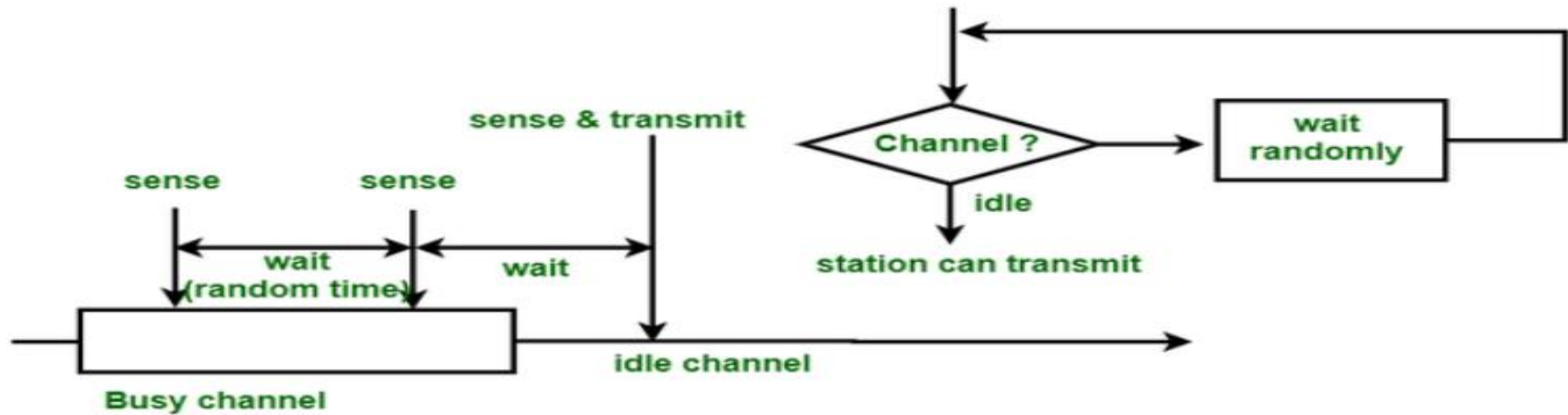
Key Points :

- Balanced approach between speed and collision avoidance.
- Reduces the chance of collisions compared to 1-persistent CSMA.
- Useful when time-slotted communication is in place.

p-persistent CSMA (Carrier Sense Multiple Access)

It applies to slotted channel and works as follows: When a station becomes ready to send, it sense the channel. If it is idle, it transmits with a probability p . With a probability $q=1-p$, it defers until the next slot. If that slot is also idle, it either transmits or defers again, with probabilities p and q . *This process is repeated until either the frame has been transmitted or another station has begun transmitting.*

p-persistent CSMA (Carrier Sense Multiple Access)



Key Points:

- Focuses on collision avoidance over speed.
- Suitable for systems where minimizing collisions is more important than transmission speed.

Parameter	1-persistent CSMA	p-persistent CSMA	Non-persistent CSMA
Carrier Sense	It sends with the probability of 1 when channel is idle.	It sends with the probability of p when channel is idle.	It send when channel is idle.
Waiting	It continuously senses the channel or carrier.	It waits for the next time slot.	It will wait for the random amount of time to check the carrier.
Chances of Collision	There is highest chances of collision in this.	Less chances as compared to 1-persistence and non-persistence.	Less chances as compared to 1-persistence but more than the p-persistence.
Utilization	It's utilization is above <u>ALOHA</u> as <u>frames</u> are only sent when the channel is idle.	It's utilization is depend upon the probability p.	It's utilization is above 1-persistent as not all the stations constantly check the channel at the same time.
Delay Low Load	It is low as frames are sent when the channel become idle.	It is large when p is small as station will not always send when channel is idle.	It is small as station will send whenever channel is found idle but longer than 1-persistent since it checks for the random time when busy.
Delay High Load	It is high due to <u>collision</u> .	It is large when the probability p of sending is small when channel is idle and channel is rarely idle.	It is longer than 1-persistent as channel is checked randomly when busy.

CSMA with Collision Detection(1)

Another improvement over ALOHA is for station to abort their transmissions as soon as they detect a collision. In other words, if two stations sense the channel to be idle and begin transmitting simultaneously, they will both detect the collision almost immediately. Rather than finish transmitting their frames, which are irretrievably garbled anyway, they should abruptly stop transmitting as soon as the collision is detected. Quickly terminating damaged frames saves time and bandwidth.

This protocol is widely used on LANs in the MAC sublayer.

CSMA with Collision Detection(1)

1. The Core Concept

"Listen Before Talk": CSMA/CD is an improvement over older protocols (like ALOHA) because a station will always "sense" (listen to) the cable before sending data to ensure it isn't already busy.

"Stop on Crash": Even if two stations listen, hear silence, and send at the exact same time, a collision will happen. CSMA/CD stations continuously listen *while* transmitting. If they detect a collision, they instantly abort the transmission rather than wasting time sending the rest of a corrupted frame.

CSMA with Collision Detection(1)

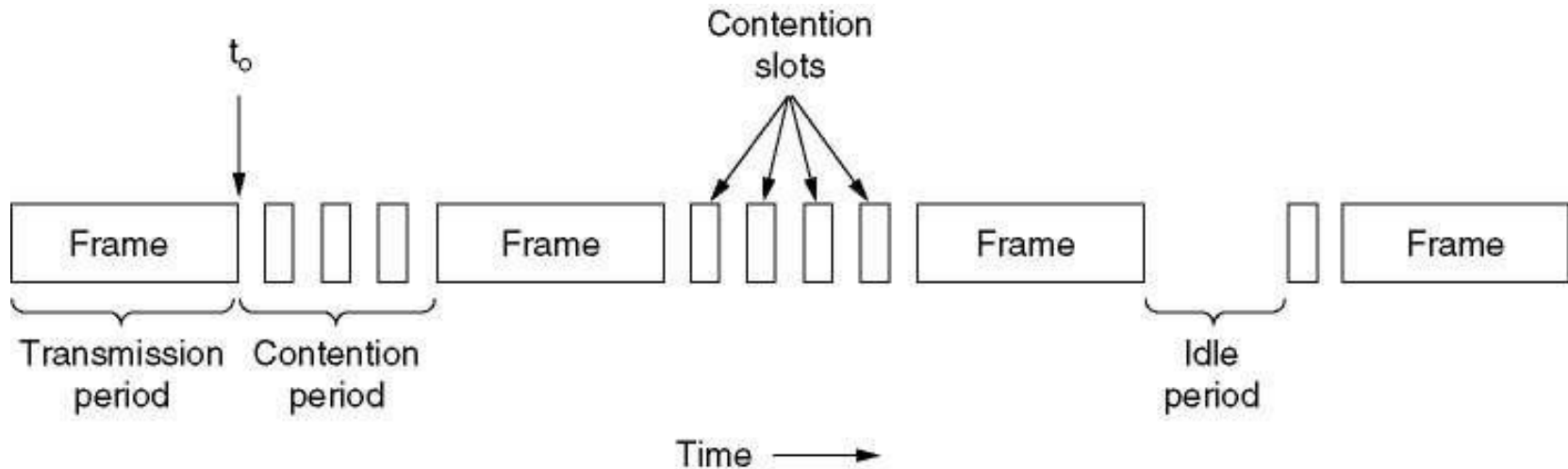
2. Hardware Requirements

- **Analog Detection:** Detecting a collision is an analog process. The hardware reads the voltage/signal on the wire while sending. If the signal it reads is different from the signal it is generating, it knows a collision occurred.
- **Not Great for Wireless:** This protocol relies on the received signal being strong enough to compare against the transmitted signal. In wireless networks, incoming signals are drastically weaker (often 1,000,000 times weaker) than outgoing ones, making collision detection highly difficult..

CSMA with Collision Detection(1)

3. Network States The protocol loops through three phases:

- Idle:** The network is completely quiet.
- Contention:** Stations attempt to grab the channel. This is the period where collisions can happen.
- Transmission:** A station has successfully seized the channel and is sending its frame without interference.



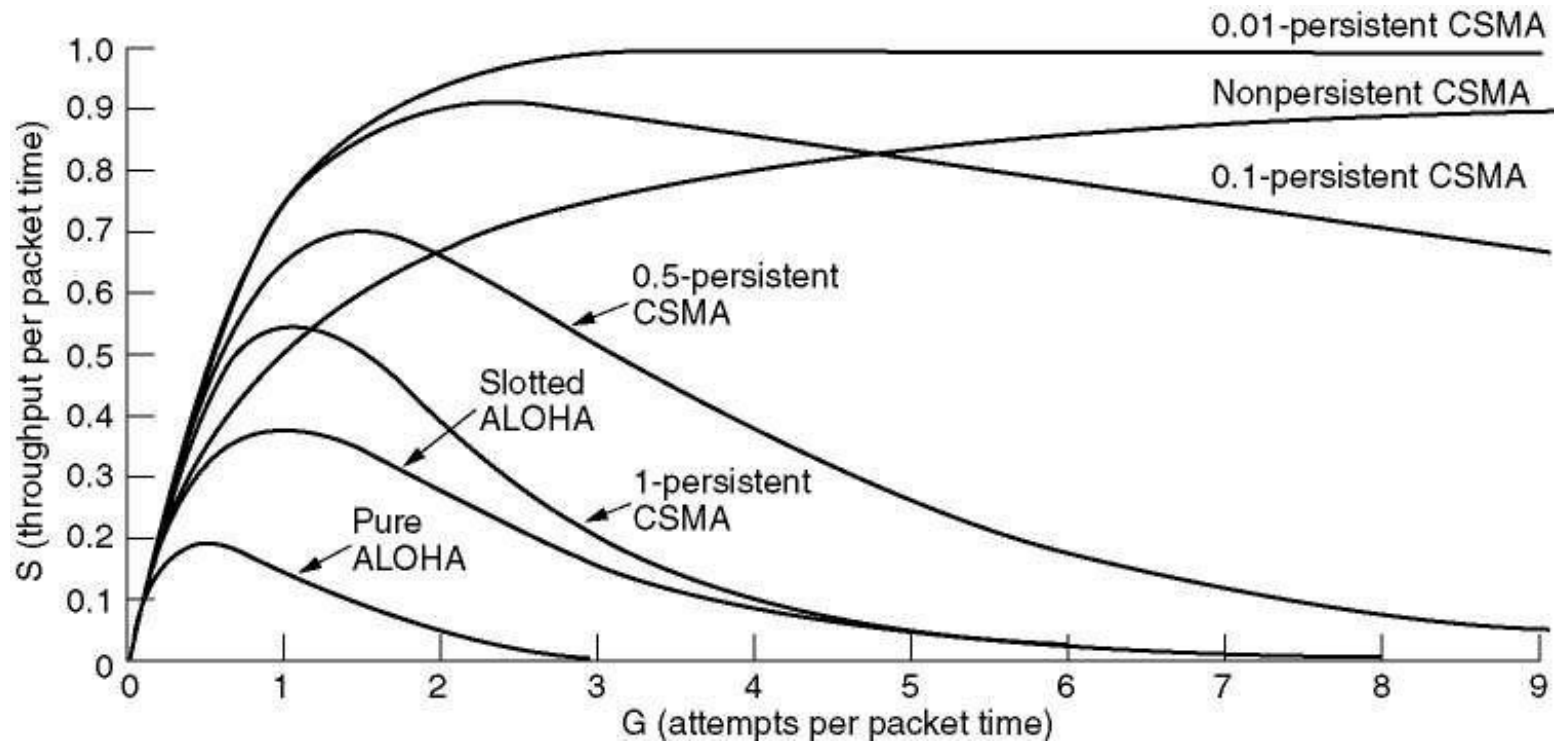
CSMA/CD can be in one of three states: contention, transmission, or idle.

CSMA with Collision Detection(1)

4. The 2τ (Two-Tau) Rule: The Worst-Case Scenario To guarantee that a station detects a collision, it must transmit for a minimum amount of time.

- Let τ (**Tau**) be the time it takes a signal to travel from one end of the cable to the other.
- Imagine Station A starts sending. At the very last microsecond before Station A's signal reaches the other end of the cable ($\tau - \epsilon$), Station B listens, hears silence, and starts transmitting.
- A collision happens immediately at Station B's end. However, the "noise burst" of that collision takes another full τ to travel all the way back to Station A.
- **Conclusion:** Station A will not know a collision happened until 2τ time has passed. Therefore, a station must transmit a frame long enough to take at least 2τ to send. If a frame is too small, the station finishes sending and stops listening before the collision noise makes it back.

CSMA PROTOCOLS' PERFORMANCE



Comparison of the channel utilization versus load for various random access protocols.

Problem: Consider a classic Ethernet LAN using CSMA/CD over a 1-kilometer (1000 meters) coaxial cable. The network bandwidth (data rate) is 10 Mbps (Megabits per second). The speed of the signal propagating through the copper cable is 2×10^8 meters/second.

1. What is the worst-case collision detection time (the contention slot width)?
2. What is the absolute minimum frame size (in bits) a station must transmit to ensure it can detect a worst-case collision?

Step 1: Calculate the Propagation Delay (τ)

The propagation delay is the time it takes a bit to travel from one end of the network to the other.

- Formula: $\tau = \frac{\text{Distance}}{\text{Speed}}$
- $\tau = \frac{1000 \text{ m}}{2 \times 10^8 \text{ m/s}}$
- $\tau = 5 \times 10^{-6}$ seconds (or $5\mu\text{s}$) (Note: This perfectly matches the text's mention of " $\sim 5 \mu\text{sec}$ " for a 1-km cable).

Step 2: Calculate the Worst-Case Collision Window (2τ)

Because the collision noise has to travel back to the sender, we double the propagation delay.

- $2\tau = 2 \times 5\mu\text{s}$
- $2\tau = 10\mu\text{s}$ (or 10×10^{-6} seconds)

Step 3: Calculate the Minimum Frame Size (L)

To guarantee collision detection, the transmission time of the frame (T_x) must be greater than or equal to the Round Trip Time (2τ).

- Formula: $T_x = \frac{\text{Length of Frame (bits)}}{\text{Bandwidth (bps)}}$
- Set $T_x \geq 2\tau$
- $\frac{L}{10,000,000 \text{ bps}} \geq 10 \times 10^{-6} \text{ seconds}$
- $L \geq (10 \times 10^{-6}) \times (10 \times 10^6)$
- $L \geq 100 \text{ bits}$

Answer: The contention slot width is $10\mu\text{s}$, meaning any collision will be detected within this timeframe. To ensure the sender is still listening when the collision noise returns, the station must send a frame that is at least 100 bits long.

4.3 Ethernet

Ethernet is the standard, most widely used technology for connecting computers and other devices in a wired Local Area Network (LAN) or Wide Area Network (WAN). It provides a set of rules (a protocol) that allows devices to communicate with each other and share data over a physical cable.

Here is a breakdown of its key characteristics:

The Physical Connection: Unlike Wi-Fi, which uses radio waves to transmit data wirelessly, Ethernet requires a physical cable. This is typically a twisted-pair copper cable (like the familiar Cat5e or Cat6 cables with RJ45 connectors) or, in high-speed enterprise networks, fiber optic cables.

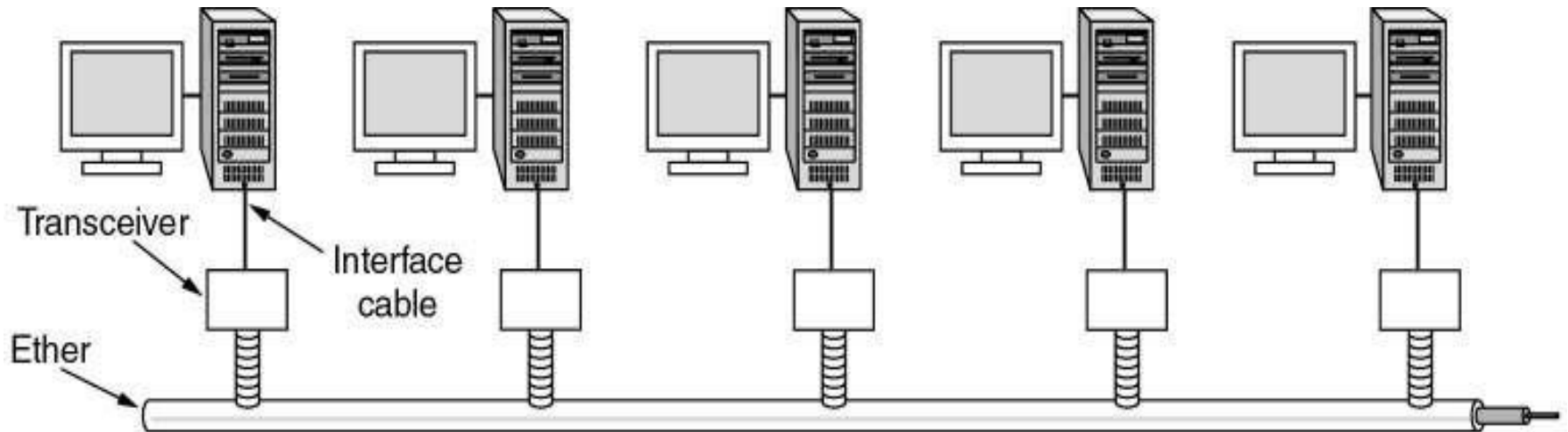
How It Works (Layer 2): Ethernet operates primarily at the Data Link Layer of the network. It takes data, breaks it down into chunks called "frames," and uses **MAC addresses** (unique hardware identifiers burned into every network card) to ensure those frames are delivered to the correct physical device on the local network.

4.3 Ethernet

Speed and Evolution: It is standardized by the IEEE as the **802.3 standard**. While early Ethernet networks in the 1980s ran at a speed of 10 Megabits per second (Mbps), modern versions like Gigabit Ethernet (1 Gbps) and 10-Gigabit Ethernet (10 Gbps) are incredibly fast. Data centers even use 400+ Gbps Ethernet.

Traffic Management: Historically, older Ethernet setups used a rule called **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) to "listen" to the cable and prevent devices from talking over each other. Modern Ethernet networks use "switches" that create dedicated, full-duplex paths between devices, making collisions a thing of the past.

Ethernet



a) Architecture of the original Ethernet.

The IEEE has standardized a number of local area networks and metropolitan area networks under the name of IEEE 802

IEEE 802 Standards

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

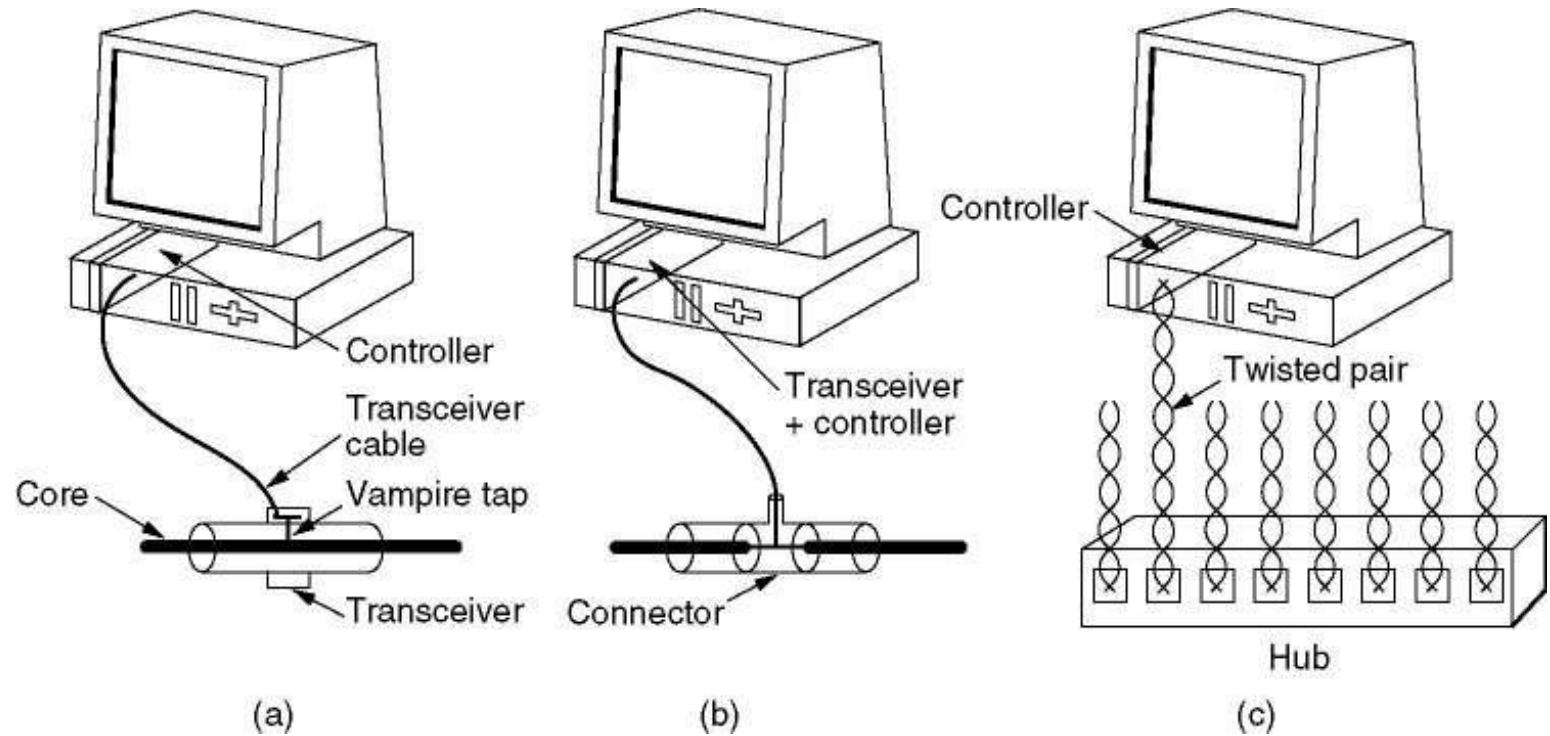
The 802 working groups. The important ones are marked with *. The ones marked with ↓ are hibernating. The one marked with † gave up.

Ethernet Cabling

Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings

The most common kinds of Ethernet cabling.

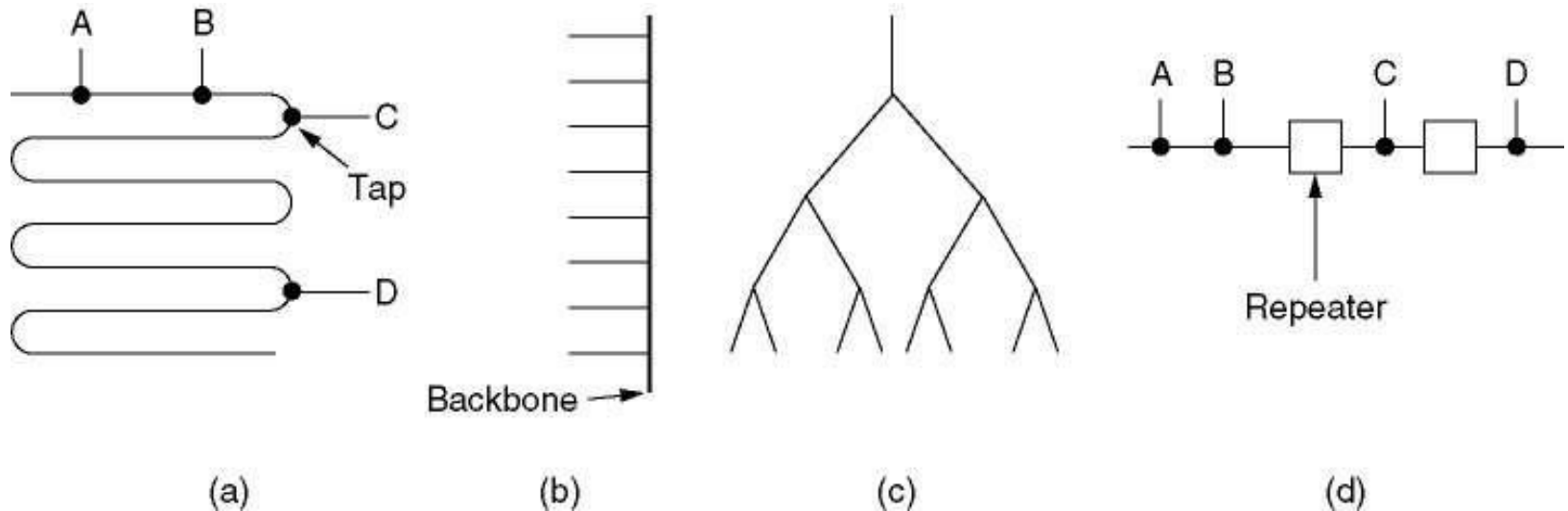
Ethernet Cabling (2)



Three kinds of Ethernet cabling.

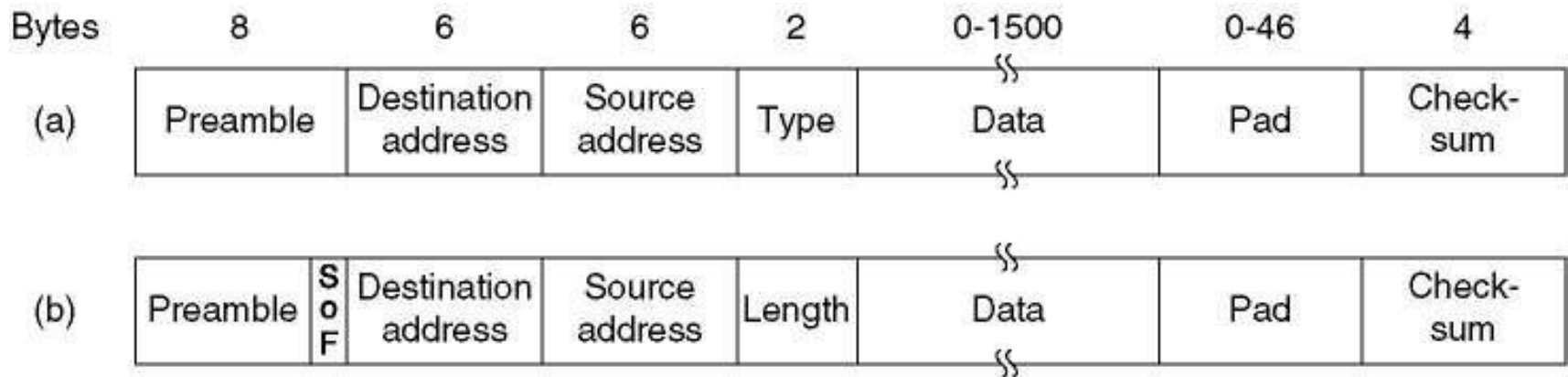
(a) 10Base5, (b) 10Base2, (c) 10Base-T.

Ethernet Cabling (3)



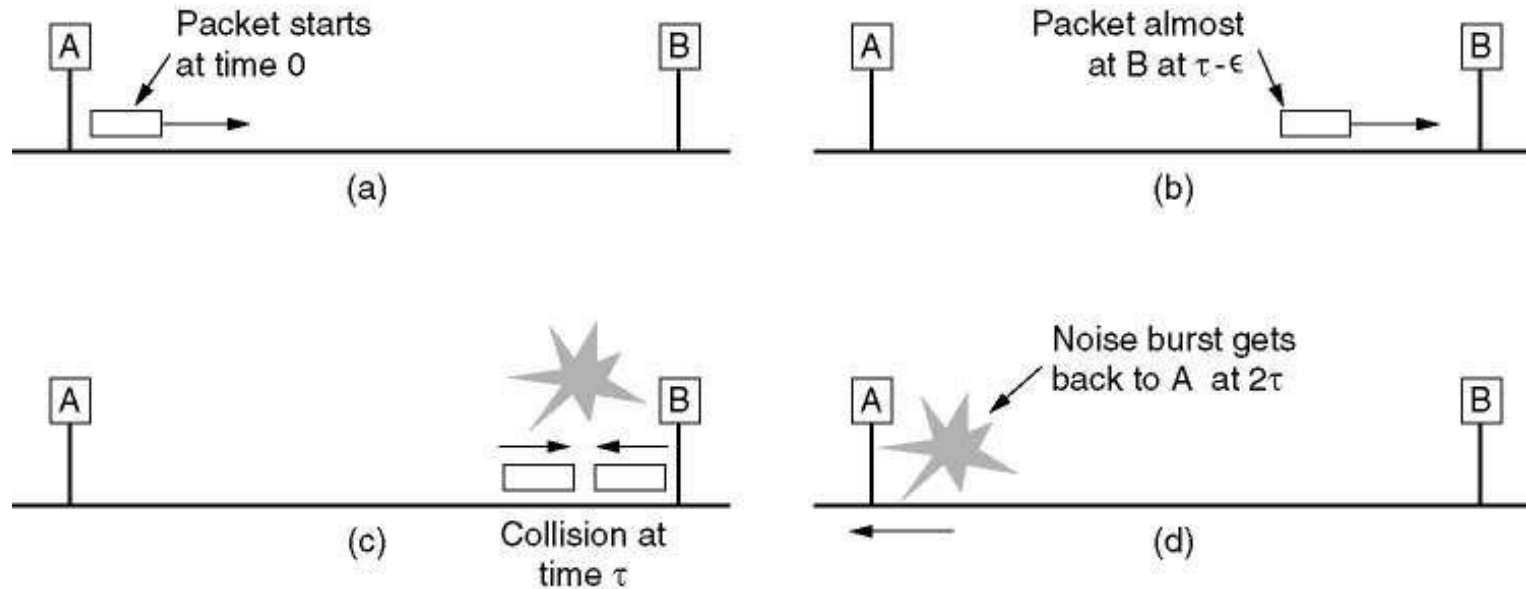
Cable topologies. (a) Linear, (b) Spine, (c) Tree, (d) Segmented.

Ethernet MAC Sublayer Protocol



Frame formats. (a) DIX Ethernet, (b) IEEE 802.3.

Ethernet MAC Sublayer Protocol (2)



Collision detection can take as long as 2τ .

MAC Addressing

Media Access Control (MAC) address is a physical address that works at the data link layer of the OSI model.

- A MAC address is a 48 or 64-bit address associated with a network adapter.
- MAC addresses are linked to the hardware of the network adapters, hence they are also known as the "hardware address" or "physical address."
- MAC addresses uniquely identify the adapter on the LAN.
- MAC addresses are expressed in hexadecimal notation. For example, "01-23-45-67-89-AB" in a 48-bit address or "01-23-45-67-89-AB-CD-EF" in a 64-bit address. Sometimes, colons (:) are used instead of dashes (-).
- MAC addresses are often considered permanent, but in some conditions, they can be changed.

Types of MAC Addresses

There are three types of MAC addresses

- Unicast MAC Address
- Multicast MAC address
- Broadcast MAC address

Unicast MAC Address: A Unicast MAC address is used for one-to-one communication.

Data is sent from one sender to one specific receiver

Each device on a network has a unique MAC address

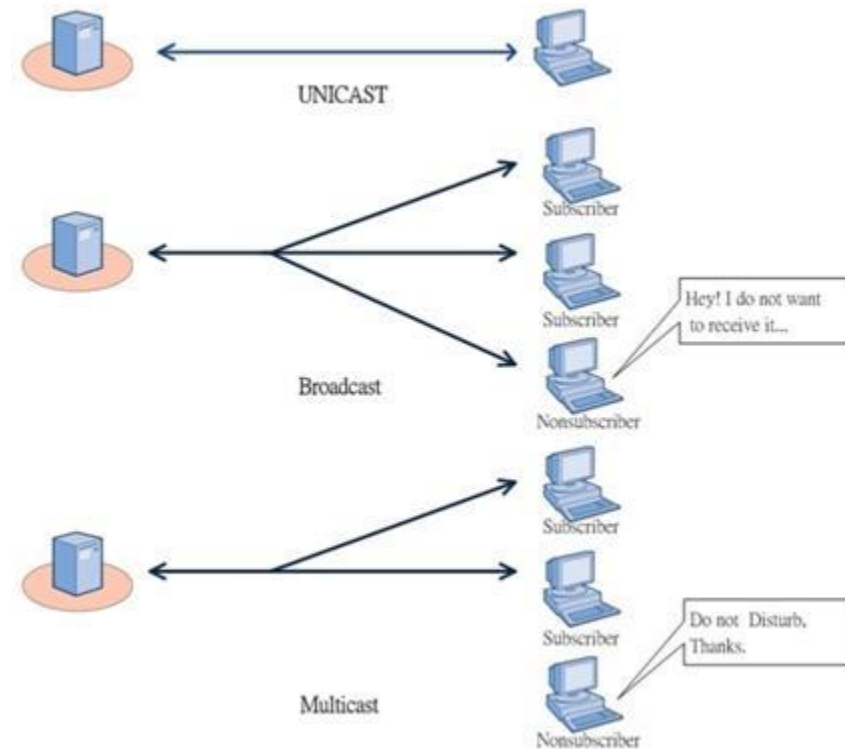
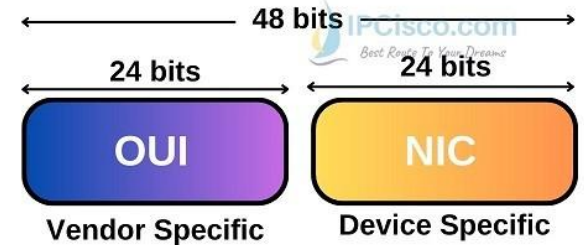
The frame is delivered only to the intended device

Example:

A computer sending data to a printer in the same network.

MAC Address

A1 : B4 : C5 : C1 : DD : 3E



A **Multicast MAC address** is used for one-to-many communication (selected group).

- Data is sent to a group of devices, not all
- Only devices that are part of the multicast group receive the data
- Common in applications like video streaming or conferencing
- Example:

Streaming a live lecture to multiple students who joined a group.

A **Broadcast MAC address** is used for one-to-all communication.

- Data is sent to all devices in the network
- The broadcast MAC address is:
- FF:FF:FF:FF:FF:FF
- Every device in the network receives and processes the frame

Example:

ARP request: "Who has this IP address?"

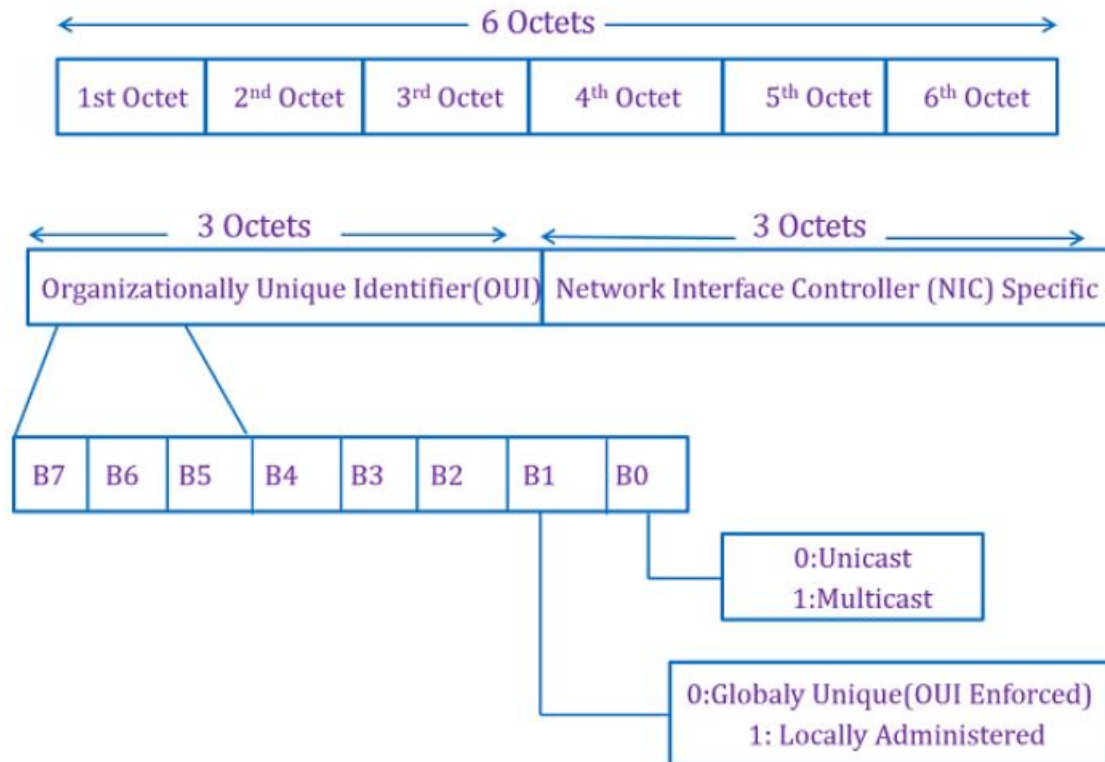
MAC Address Format

A 48-bit MAC address is represented as a string of six octets,

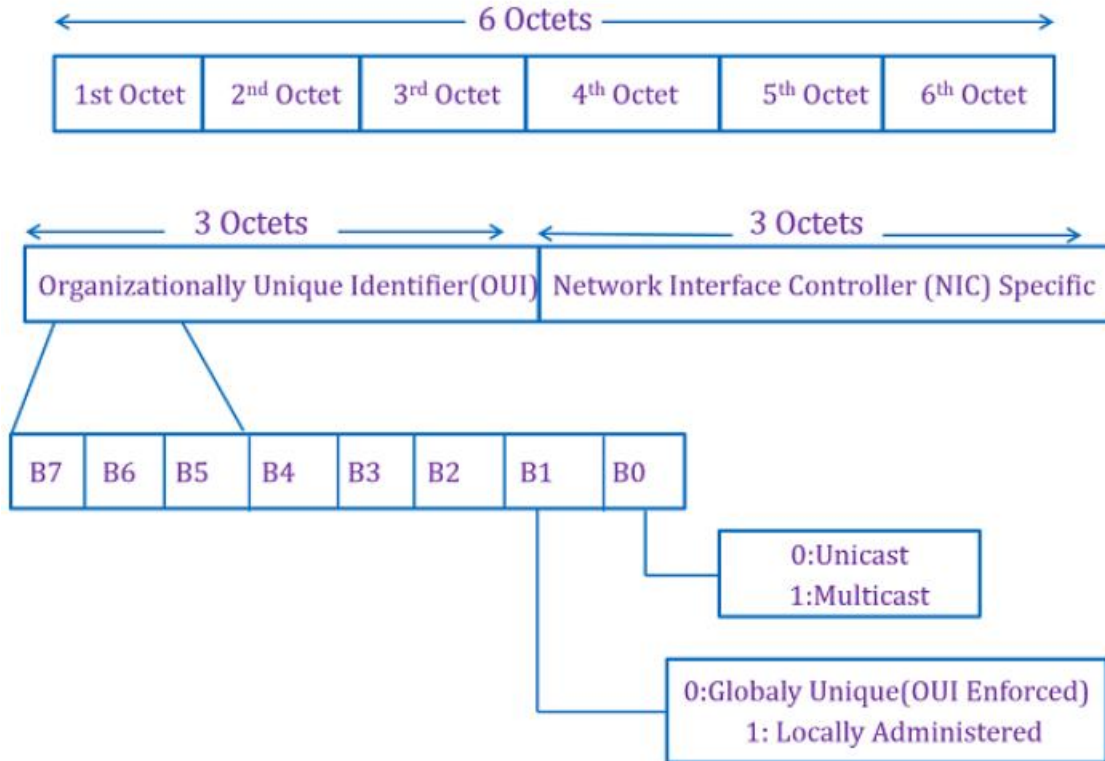
"MM:MM:MM:SS:SS:SS".

The first half (24 bits) of the MAC address contains the ID number of the adapter manufacturer. These IDs are regulated by an Internet standards organization.

The second half (24 more bits) of the MAC address represents the serial number assigned to the adapter by the manufacturer.



MAC Address Format



OUIs of some well-known manufacturers

- "00-14-22" Dell
- "00-04-DC" Nortel
- "3C:5A: B4" Google, Inc.